

Supplementary Materials:

SymC Power Grid Optimization: Harnessing Scale-Invariance and Substrate-Inheritance Alignment for Predictive Infrastructure Stability and Control

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1 Extended Theoretical Derivations

1.1 Complete Swing Equation Derivation

The electromechanical swing equation connects mechanical and electrical dynamics. Starting from rotational mechanics:

$$J \frac{d^2 \theta_m}{dt^2} = T_m - T_e - D_m \frac{d\theta_m}{dt}, \quad (1)$$

where J is moment of inertia, θ_m mechanical angle, T_m mechanical torque (prime mover), T_e electrical torque (load), D_m mechanical damping.

Converting to electrical angle $\delta = (p/2)\theta_m$ where p is number of poles, and normalizing by base power S_{base} :

$$\frac{2H}{\omega_s} \frac{d^2 \delta}{dt^2} = P_m - P_e - D \frac{d\delta}{dt}, \quad (2)$$

where $H = J\omega_s^2/(2S_{\text{base}})$ is inertia constant (seconds), P_m mechanical power, P_e electrical power, ω_s synchronous speed.

For small deviations $\Delta\delta$ from equilibrium δ_0 :

$$\frac{2H}{\omega_s} \Delta\ddot{\delta} + D\Delta\dot{\delta} + P_s\Delta\delta = 0. \quad (3)$$

Dividing by the coefficient of $\Delta\ddot{\delta}$:

$$\Delta\ddot{\delta} + \frac{D\omega_s}{2H} \Delta\dot{\delta} + \frac{P_s\omega_s}{2H} \Delta\delta = 0, \quad (4)$$

which is the standard damped harmonic oscillator form used throughout the main paper.

Natural frequency:

$$\omega_n = \sqrt{\frac{P_s\omega_s}{2H}}. \quad (5)$$

Damping ratio:

$$\chi = \frac{D}{2\sqrt{MK}} = \frac{D\omega_s}{4H\omega_n}. \quad (6)$$

1.2 Participation Factors

Mode shapes (eigenvectors) ϕ_i indicate spatial distribution of each mode. The participation factor for generator k in mode i is:

$$p_{i,k} = \frac{|\phi_{i,k}|^2}{\sum_j |\phi_{i,j}|^2}, \quad (7)$$

normalized such that $\sum_k p_{i,k} = 1$.

Regional damping ratio for area S is computed as:

$$\chi(S) = \sum_i w_i p_{i,S} \chi_i, \quad (8)$$

where w_i weights modes by energy content and $p_{i,S} = \sum_{k \in S} p_{i,k}$ aggregates participation over all generators in region S .

1.3 Exceptional Point Mathematics

At the exceptional point where $\chi = 1$, the discriminant vanishes. For $\lambda^2 + 2\chi\omega_n\lambda + \omega_n^2 = 0$:

Discriminant:

$$\Delta = 4\chi^2\omega_n^2 - 4\omega_n^2 = 4\omega_n^2(\chi^2 - 1). \quad (9)$$

At $\chi = 1$: $\Delta = 0$, yielding repeated real root $\lambda = -\omega_n$.

Sensitivity divergence:

$$\frac{\partial \lambda}{\partial \chi} = -\omega_n \pm \frac{\chi\omega_n}{\sqrt{\chi^2 - 1}}. \quad (10)$$

As $\chi \rightarrow 1$:

$$\left| \frac{\partial \lambda}{\partial \chi} \right| \sim \frac{\omega_n}{|\chi - 1|^{1/2}} \rightarrow \infty. \quad (11)$$

This divergence is signature of exceptional points where eigenvalues coalesce.

2 Detailed Methodology

2.1 Chi Extraction Pipeline

Event detection triggers on frequency derivative $|df/dt| > 0.05$ Hz/s. For each 30 s ringdown window, Prony analysis constructs a Hankel matrix (model order $L = 15$, validated via AIC) from detrended frequency measurements. Generalized eigenvalue decomposition extracts modal frequencies ω_i and damping rates σ_i . For oscillatory modes, $\chi_i = \sigma_i/|\lambda_i|$; for aperiodic modes, $\chi = -(\lambda_1 + \lambda_2)/(2\sqrt{\lambda_1\lambda_2})$. Regional χ aggregates modes weighted by participation factors and modal energy.

Quality control: Modes accepted if residual error $\varepsilon < 0.05$ and SNR > 20 dB. Bootstrap confidence intervals (1000 resamples) quantify uncertainty. Cross-timescale divergence $\Delta\chi(T_1, T_2)$ computed via band-pass filtering: fast (0.1–2 Hz), medium (0.01–0.1 Hz), slow (0.001–0.01 Hz). Threshold $|\Delta\chi| > 0.2$ sustained > 5 min optimized via ROC analysis (Youden's $J = 0.87$).

2.2 Simulation Protocol Specification

Multi-session analysis: 50 independent runs, each 20–30 hours simulated time. Events randomly scheduled (mean inter-arrival 3 hours, exponentially distributed) with types: generator trips (40%), line outages (35%), load shocks (25%). Event magnitudes: $\Delta P \in [50, 1500]$ MW uniform random.

Lifecycle simulation: Single 83-hour run with 6 major events ($\Delta P > 1000$ MW) scheduled at $t = 1200, 3500, 5200, 6800, 7400, 8900$ minutes to track cumulative substrate degradation. Initial conditions: IEEE 39-bus at nominal load, all generators online, $\chi_0 = 1.00 \pm 0.02$. Control settings: standard PSS gains, AGC 5-minute time constant, no special compensators.

Validation: All simulations use identical baseline parameters but independent random event sequences. Across 50 runs: $\mu_\chi^{(\text{stable})} = 0.82$, $\sigma_\chi^{(\text{stable})} = 0.11$; $\mu_\chi^{(\text{event})} = 0.43$, $\sigma_\chi^{(\text{event})} = 0.18$, yielding $t = 8.7$, $p < 0.001$.

2.3 Statistical Analysis Protocol

Confidence intervals: Bootstrap (1000 resamples) for correlation coefficients and mean differences. Reported 95% CI derived from 2.5th and 97.5th percentiles.

Threshold selection: ROC analysis on 50-simulation training set, validated on independent UK field data (2,847 snapshots, 47 events). Youden’s J maximized at $\chi_{\text{threshold}} = 0.60$ for early warning crossing.

Multiple comparisons: Reported p-values are uncorrected; Bonferroni adjustment would maintain significance for key claims (stable vs. event χ difference: $p < 0.001/6$ even after correction).

Effect sizes: Stable vs. event distributions show Cohen’s $d = 2.3$ (very large effect), indicating practically meaningful separation.

2.4 Two-Level Structure: Mathematical vs. Operational

The framework operates at two distinct levels:

Mathematical level: The exceptional point at $\chi = 1$ represents the precise boundary where eigenvalues coalesce. This is exact, substrate-independent, and universal across all systems governed by second-order dynamics.

Operational level: Real grids operate at $\chi \approx 0.8\text{--}0.9$, maintaining margins below the mathematical boundary due to: (1) measurement uncertainty ($\pm 0.03\text{--}0.05$), (2) control delays (100–500 ms), (3) stochastic disturbances (ambient noise), (4) risk-averse operational practice. The 0.1–0.2 gap between operational optimum and mathematical boundary represents the “safety margin”.

This two-level structure explains why numerical simulations reach χ values far from operational clustering. In simulation, perfect knowledge and instantaneous control are imposed. Real systems necessarily operate with margins. Both behaviors are consistent with the same underlying principle: stability is determined by proximity to the $\chi = 1$ boundary.

Empirical prediction: Systems with better instrumentation, faster control, and lower uncertainty should cluster closer to $\chi = 1.0$. Conversely, systems with legacy control and high noise should cluster lower (0.70–0.80). This provides testable prediction for future cross-system comparisons.

3 Grid Stability Space and Failure Taxonomy

3.1 The (χ , T, S) Coordinate System

χ axis (Boundary Position):

- Safe zone: $0.7 \leq \chi \leq 1.0$ (adaptive window)
- Yellow: $0.5 \leq \chi < 0.7$ (underdamped, elevated risk)
- Red: $\chi < 0.5$ (severely underdamped)

T axis (Time/Inertia):

- Safe: $H > 3$ s (adequate buffer)
- Yellow: $1 < H < 3$ s (reduced margin)
- Red: $H < 1$ s (massless, RoCoF concerns)

S axis (Structure/Stiffness):

- Safe: $\text{SCR} > 3$ (strong grid)
- Yellow: $2 < \text{SCR} < 3$ (weak grid)
- Red: $\text{SCR} < 2$ (very weak)

3.2 Failure Mode Classification

3.2.1 Type 1: Stiffness Collapse

SymC diagnosis: Loss of structural integrity, $P_s \rightarrow 0$, $\text{SCR} < 1.5$

Mechanism: As reactive support depletes:

$$P_s = \frac{|V_1||V_2|}{X} \cos \delta \quad (12)$$

decreases. Near collapse: $V_2 \rightarrow V_{\min}$, driving:

$$\omega_n = \sqrt{\frac{P_s \omega_s}{2H}} \rightarrow 0. \quad (13)$$

With finite D :

$$\chi = \frac{D\omega_s}{4H\omega_n} \rightarrow \infty. \quad (14)$$

3.2.2 Type 2: Controller Failure

SymC diagnosis: Loss of damping, $\chi < 0$ or high σ_χ

Mechanism: High-gain controllers introduce negative damping:

$$D_{\text{total}} = D_{\text{natural}} + D_{\text{PSS}} + D_{\text{AVR}}, \quad (15)$$

where $D_{\text{AVR}} < 0$. When $D_{\text{total}} < 0$:

$$\chi < 0. \quad (16)$$

Hopf bifurcation to limit cycle—sustained oscillations.

3.2.3 Type 3: Impedance Mismatch

SymC diagnosis: Interface reflection, $\Delta Z \neq 0$

Mechanism: Series compensation shifts electrical resonance:

$$f_{\text{er}} = \frac{1}{2\pi\sqrt{LC}}, \quad f_{\text{sub}} = f_0 - f_{\text{er}}. \quad (17)$$

If f_{sub} matches mechanical torsional frequency, $D_{\text{total}}(f_{\text{sub}}) \approx 0$, driving $\chi_{\text{mech}} \rightarrow 0$ at subsynchronous frequency.

4 Inverter Physics and Control

4.1 Phase Transition to Massless Substrate

Synchronous generators store kinetic energy:

$$E_{\text{kinetic}} = \frac{1}{2} J \omega^2. \quad (18)$$

Aggregate inertia:

$$H_{\text{sys}} = \frac{\sum_i H_i S_i}{\sum_i S_i}. \quad (19)$$

As IBR penetration $\rightarrow 100\%$: $H_{\text{sys}} \rightarrow 0$. RoCoF:

$$\frac{df}{dt} = \frac{\Delta P}{2H_{\text{sys}} \cdot S_{\text{base}}} \rightarrow \infty. \quad (20)$$

Grid becomes massless substrate—no temporal ballast.

4.2 Virtual Synchronous Machine Control

GFM inverters emulate swing equation:

$$J_{\text{virt}} \frac{d\omega}{dt} = P_{\text{ref}} - P_{\text{meas}} - D_{\text{virt}}(\omega - \omega_0), \quad (21)$$

$$\frac{d\delta}{dt} = \omega - \omega_0. \quad (22)$$

Output angle: $\theta(t) = \omega_0 t + \delta(t)$.

Power injection:

$$P_{\text{inv}} = P_{\text{ref}} - D_{\text{virt}}\Delta\omega + K_p\Delta f. \quad (23)$$

4.3 Critical Emulation Principle

To maintain $\chi = 1$ dynamically:

$$D_{\text{virt}}(t) = 2\sqrt{J_{\text{virt}} \cdot P_s(t)}. \quad (24)$$

5 Extended Empirical Results

5.1 Pristine Substrate Recovery

Baseline for pristine substrate behavior: System experiences 15+ disturbances but fully recovers to adaptive window within 20 minutes. No persistent baseline shift.

Recovery time constants:

- Minor events ($\Delta\chi < 0.2$): $\tau = 8 \pm 3$ min
- Moderate events ($0.2 \leq \Delta\chi < 0.5$): $\tau = 14 \pm 4$ min
- Major events ($\Delta\chi > 0.5$): $\tau = 18 \pm 5$ min

Statistical characterization:

- Baseline mean: $\chi_{\text{baseline}} = 1.00 \pm 0.02$ throughout
- Variance: $\sigma_\chi = 0.03$ (stable, low volatility)
- Trend: Linear regression $\beta = -0.0002$ per min, $p = 0.73$ (not significant)
- No baseline drift—confirms pristine substrate property

5.2 Real-World Validation Maps: Multi-Domain Clustering

The following maps show χ clustering across real operational data from distinct grid topologies and sizes. All points are measured from actual synchrophasor data during normal and stressed operating conditions.

6 Falsification Protocols

6.1 Prediction 1: Eigenvalue Coalescence at $\chi = 1$

Test procedure:

1. Sweep damping parameter D in simulation
2. Extract eigenvalue spectrum at each D value
3. Plot $\text{Im}(\lambda)$ vs. $\text{Re}(\lambda)$ as D varies

4. Monitor discriminant $\Delta = 4\omega_n^2(\chi^2 - 1)$

Success criterion: Complex conjugate eigenvalues merge to real axis when $\chi = 1 \pm 0.02$.

Timeline: Laboratory validation 2025–2026 using real-time digital simulator (RTDS).

6.2 Prediction 2: Adaptive Window Clustering

Test procedure:

1. Collect $\chi(t)$ for 10,000+ snapshots across multiple grids
2. Histogram χ for stable vs. event conditions
3. Statistical comparison (t-test, Kolmogorov-Smirnov)
4. ROC analysis for binary classification

Success criterion: Mean $\chi_{\text{stable}} - \chi_{\text{event}} > 0.2$ with $p \leq 0.01$.

Status: VALIDATED (main paper). Ongoing extension to additional grids.

6.3 Prediction 3: Cross-Timescale Divergence as Early Warning

Test procedure:

1. Compute $\Delta\chi(T_{\text{fast}}, T_{\text{slow}})$ continuously
2. Set threshold: $|\Delta\chi| > 0.2$ sustained > 5 min
3. Label ground truth: failure within next 2 hours (yes/no)
4. Construct confusion matrix, compute sensitivity/specificity

Success criterion: Sensitivity $> 80\%$, specificity $> 95\%$, lead time > 60 min.

Status: VALIDATED (89% sensitivity, 98% specificity, 127 min lead time).

6.4 Prediction 4: Substrate Hysteresis

Test procedure:

1. Monitor χ_{baseline} before and after major events
2. Track recovery over extended period (hours to days)
3. Test null hypothesis: $\chi_{\text{post}} = \chi_{\text{pre}}$
4. Measure permanent offset: $\Delta\chi_{\text{permanent}}$

Success criterion: Statistically significant baseline shift ($\Delta\chi > 0.01$, $p \leq 0.05$) persisting > 1 hour post-event.

Status: VALIDATED. Permanent shifts of 1–3% observed.

7 Limitations and Future Work

7.1 Linearization Regime

χ extraction requires local linearization around operating points. Highly nonlinear phenomena (voltage collapse post-bifurcation, pole slipping) fall outside small-signal approximation. SymC provides early warning of approach to nonlinear boundaries but does not model post-instability dynamics.

Valid regime: $\chi > 0.1$. Below this, nonlinear methods required.

7.2 Modal Observability

Accurate $\chi(T, S)$ field construction requires:

- Sufficient PMU/FNET coverage ($> 80\%$ of major nodes)
- Mode excitation (ambient noise or disturbances)
- $\text{SNR} > 20$ dB for reliable extraction

Weakly observable modes or sparse telemetry exhibit higher uncertainty (± 0.08 vs. ± 0.03).

7.3 Stochastic Effects

Framework does not currently model explicit stochastic noise processes. Extensions to stochastic differential equations are straightforward but not implemented.

While variance σ_χ serves as indicator, explicit probabilistic modeling would enhance risk quantification.

7.4 Complementarity with Existing Tools

Critical-damping framework provides real-time stability assessment but does not replace:

- EMT simulation for protection coordination
- Detailed inverter controller validation
- Subsynchronous interaction studies
- Long-term planning tools

Framework is monitoring/predictive layer, not planning tool replacement.

7.5 Future Extensions

Very fast timescales (< 100 ms): Electromagnetic transients, protection dynamics. Requires higher sampling rates.

Very slow processes (> 1 week): Infrastructure aging, seasonal patterns. Requires long-term data collection.

Distributed energy resources: Aggregated behind-the-meter resources. Requires distribution-level monitoring.

Multi-area coordination: Cross-border stability assessment. Requires data sharing and standardized protocols.